

Drought or Flood Warning Decision made based on Bayesian Network

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October 6, 2025

Abstract

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1 Introduction

Drought Anticipatory Action (DAA)

bnet (Bayesian network) (see Ref.[4])

Bring the following disaster prediction models together via Damage Functions

- (a) Ref.[3] book by Tim Palmer
- (b) Ref.[6] Climada Software
- (c) Ref.[1] by Huizinga et al. Global flood depth damage functions
- (d) Ref.[2] by Liu et al. A method of deriving damage fractions functions.

2 Notation and Conventions

We will use $\mathbb{1}(S)$ to denote the indicator function (a.k.a. the truth function). $\mathbb{1}(S)$ equals 1 if statement S is true, and 0 if S is false. For example, $\mathbb{1}(x \geq 0)$ equals 0 if $x < 0$ and 1 otherwise.

In this article, random variables will be indicated by underlining. For example, $P(\underline{x} > x)$ will denote the probability that the random variable $\underline{x}$ is greater than x .

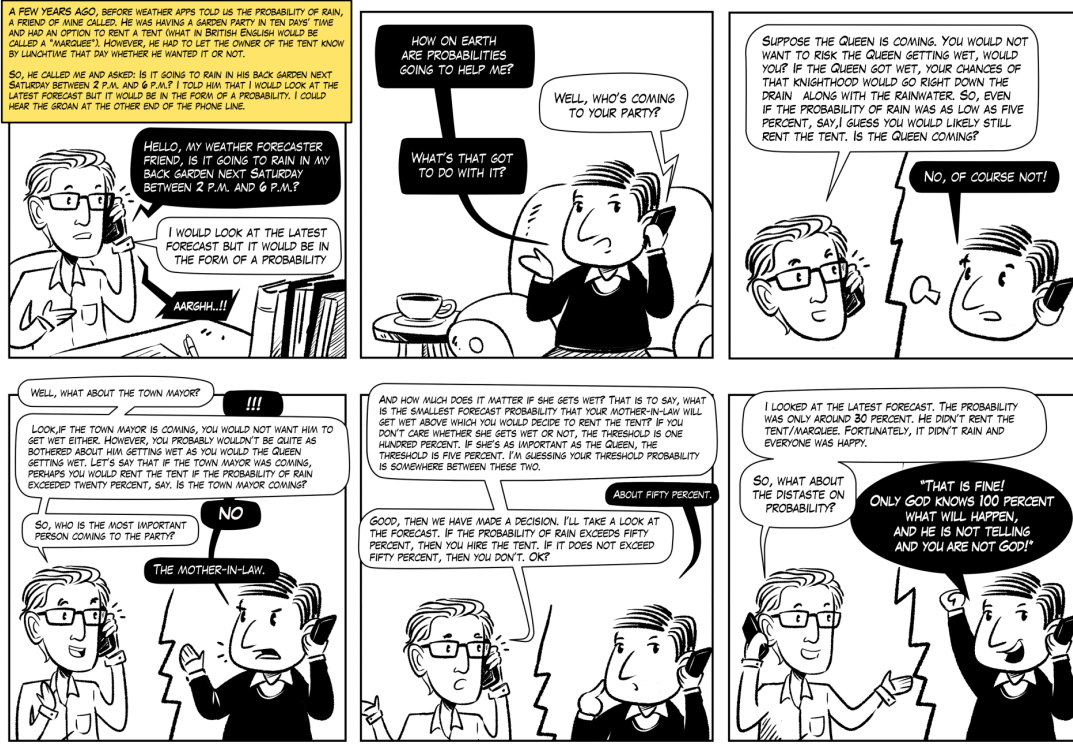


Figure 1: Illustration from Tim Palmer's book Ref.[3]

3 Model

α = index that ranges over different hazards

$f_\alpha(x_\alpha)$ = **Damage Fraction** (term used in Ref.[1]), **Impact Function** (term used in Ref.[6]) for Hazard α . $f_\alpha(x_\alpha)$ is a function of the parameter x_α , and increases monotonically from 0 when $x_\alpha = 0$ to 1 when x_α is maximum, $x_\alpha = x_{\alpha,max}$

$D_{\alpha,max}$ = **Maximum Damage** (term used in Ref.[1]), **Exposure** (term used in Ref.[6]) for Hazard α . In dollars.

Henceforth, anything called a Damage will be in units of dollars (or some other currency), except damage fractions, which will be unitless, and range from zero to one.

$D_\alpha(x_\alpha)$ = **Damage** (term used in Ref.[1]), **Impact** (term used in Ref.[6])

$$D_\alpha(x_\alpha) = D_{\alpha,max} f_\alpha(x_\alpha) \quad (1)$$

$\bar{D}_\alpha$ = **Average Damage**

$$\bar{D}_\alpha = \frac{1}{x_{\alpha,max}} \int_0^{x_{\alpha,max}} dx f_\alpha(x_\alpha) \quad (2)$$

D_{max} = **maximum total damage** that is tolerable. Any time total Damage is higher than this, always issue drought warning.

We will use $x_\alpha = \text{None}$ to indicate

$$D_\alpha(x_\alpha = \text{None}) = \overline{D}_\alpha \quad (3)$$

When $x_\alpha \neq \text{None}$, x_α should be the value of x_α for some historical event or a bin value.

ϵ_α = **error** in D_α , normally distributed with mean zero and standard deviation σ_α .

$$D'_\alpha(x_\alpha) = D_\alpha(x_\alpha) + \epsilon_\alpha \quad (4)$$

f_D = **total Damage fraction**

$$f_D = \begin{cases} 1 & \text{if } \sum_\alpha D'_\alpha(x_\alpha) > D_{max} \\ \frac{1}{D_{max}} \sum_\alpha D'_\alpha(x_\alpha) & \text{otherwise} \end{cases} \quad (5)$$

Let

$\kappa = (\sigma, \delta)$ = the drought kind, where $\sigma \in \{\text{MAM}, \text{JJAS}\}$ ¹ is the season, and $\delta \in \{\text{Moderate}, \text{Extreme}, \text{Severe}\}$ is the drought degree.

S_κ = **Severity** of drought is defined as minus SPI² (higher rain, higher SPI, less severe drought). For flood, we would use $+SPI$ instead. Severity is called **Intensity** in Ref.[6]

$P(S_\kappa)$ = likelihood, probability that a drought of severity S_κ will occur in the near future (weeks to months), based on forecasts.

μ = ensemble member index

M = number of ensemble members, 51 for *ECMWF* (European Centre for Medium-Range Weather Forecasts)

$\sigma \backslash \delta$	Moderate	Extreme	Severe
MAM	−0.55	−0.98	−0.99
JJAS	−0.41	−0.98	−0.99

Table 1: SPI threshold for Karamoja, Uganda. Cells give $-S_\kappa^*$, where $\kappa = (\sigma, \delta)$

S_κ^* = severity threshold for drought (See Table 1 for examples of S_κ^* .)

$$P(\underline{S}_\kappa \geq S_\kappa^*) = \frac{1}{M} \sum_{\mu=1}^M \mathbb{1}(S^{\kappa, \mu} \geq S_\kappa^*) \quad (6)$$

W_κ = **drought warning**, either 0 or 1

$f_{D, \kappa}$ = the total damage fraction f_D defined previously, for drought kind κ .

We will use this as a threshold for triggering a drought warning.

¹Letters here stand for the first letter of the months of the year.

²**Standardized Precipitation Index** (SPI) ranges from −4 (least rain) to 4 (most rain).

$$W_\kappa = \mathbb{1} [P(\underline{S}_\kappa \geq S_\kappa^*) > 1 - f_{D,\kappa}] \quad (7)$$

The above model can be expressed via the Bayesian Networks in Figs.2 and

3

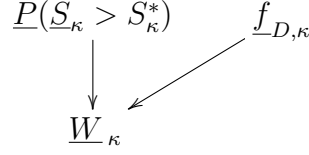


Figure 2: bnet for calculating warning $\underline{W}_\kappa$

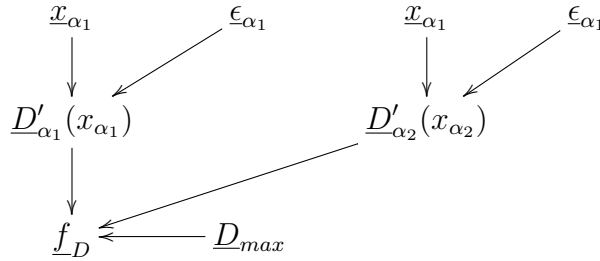


Figure 3: bnet for calculating total damage fraction f_D for two hazards α_1 and α_2 .

References

- [1] Jan Huizinga, Hans De Moel, and Wojciech Szewczyk. Global flood depth-damage functions: Methodology and the database with guidelines. https://publications.jrc.ec.europa.eu/repository/bitstream/JRC105688/global_flood_depth-damage_functions__10042017.pdf, 2017.
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